AI.408

Robot Path Planning

COURSE OVERVIEW

Computing collision-free paths: configuration spaces, sampling, optimization, and kinodynamic planning.



ILLUSTRATIVE HERO IMAGERY

1

SESSION

60 MINUTES

ORIENTATION: THE PATH PROBLEM

KEY TOPICS

- Path vs. trajectory
- Configuration space
- Obstacles as C-obstacles
- Planning queries

TEACHING FOCUS

Compute where a robot may move.

KEY TAKEAWAY Path planning finds feasible motion.

2

SESSION

60 MINUTES

CLASSICAL PLANNERS

KEY TOPICS

- Visibility graphs
- Cell decomposition
- Voronoi diagrams
- Exact cell decomposition

TEACHING FOCUS

Solve planning with geometry.

▲ **KEY TAKEAWAY** Geometry defines the free space.

3

SESSION

60 MINUTES

SAMPLING-BASED PLANNERS

KEY TOPICS

- PRM
- RRT and RRT*
- BIT*
- Informed sampling

TEACHING FOCUS

Plan in high-dimensional spaces.

KEY TAKEAWAY Sampling tames high dimensions.

4

SESSION

60 MINUTES

OPTIMIZATION-BASED PLANNING

KEY TOPICS

- Trajectory optimization
- CHMP
- Convex optimization
- Smooth paths

TEACHING FOCUS

Refine paths for quality and dynamics.

KEY TAKEAWAY Optimization polishes raw paths.

5

SESSION

60 MINUTES

KINODYNAMIC & CONSTRAINED PLANNING

KEY TOPICS

- Dynamics constraints
- Kinodynamic planning
- State lattices
- Time-parameterization

TEACHING FOCUS

Plan physically feasible motion.

KEY TAKEAWAY Feasible paths respect the robot's body.

ROBOT PATH PLANNING

6

SESSION

60 MINUTES

PATH PLANNING & REVIEW

KEY TOPICS

- Planner comparison
- Course recap and assessment
- Toward simulation (AI.409)
- Planning race

TEACHING FOCUS

Implement and compare planners.

KEY TAKEAWAY The right planner fits the robot and task.